CTD MODULE

2.4 Non Clinical Overview

Ganaxolone Oral Suspension (50 mg/mL)

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LIST OF ABBREVIATIONS

AEDs	Antiepileptic drugs
AUC	Area Under Cover
CDD	Cyclin-dependent kinase-like 5 (CDKL5) deficiency disorder
CDKL5	Cyclin-dependent kinase-like 5
ED50	Effective dose
GABA	Gamma amino butyric acid
GNX	Ganaxolone
IS	Infantile spasms
MES	Maximal electroshock
PTZ	Pentylenetetrazole
T_{max}	Maximum plasma concentration

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2.4.1 OVERVIEW OF THE NON CLINICAL TESTING STRATEGY

Seizure disorders are neurological conditions characterized by recurrent seizures that can cause significant disability and negatively impact quality of life. Among them, Cyclin-dependent kinase-like 5 (CDKL5) deficiency disorder (CDD) is a rare genetic neurodevelopmental condition characterized by early-onset, drug-resistant, and refractory seizures. CDD is recognized as one of the more frequent causes of developmental and epileptic encephalopathy.[1,2,3]

Current antiepileptic drugs often provide only partial seizure control and do not prevent the progressive changes associated with epilepsy. Neuroactive steroids have shown broad anticonvulsant activity in several animal seizure models. Ganaxolone, a synthetic neuroactive steroid, has demonstrated protection against kindled seizures in two rodent models.3

Ganaxolone, the first FDA-approved therapy for CDD-associated seizures, has also demonstrated potential benefits in treating other severe seizure disorders, including refractory epilepsy and status epilepticus.[4,5]

GNX is effective in a broad range of animal models of epilepsy and behavioral disorders, but exhibits important pharmacological differences from the benzodiazepine class of GABAA receptor modulators.[6]

A comprehensive nonclinical testing program was designed to characterize the pharmacological activity, safety profile, and pharmacokinetic properties of GNX in support of clinical development.

Literature Search Strategy

A comprehensive literature search was performed to identify nonclinical studies on the pharmacology, pharmacokinetics, pharmacodynamics, and toxicology of ganaxolone. Databases including PubMed, Embase, and Google Scholar were searched [up to November 2025] using keywords such as *ganaxolone*, *neurosteroid*, *GABA receptor modulator, preclinical*, *animal model*, and *toxicology*. Relevant peer-reviewed publications and regulatory reports were reviewed to summarize the nonclinical pharmacological and toxicological profile of ganaxolone.

2.4.2 PHARMACOLOGY

Ganaxolone is a neuroactive steroid gammaaminobutyric acid A (GABAA) receptor positive modulator. Ganaxolone (1-[(3R, 5S, 8R, 9S, 10S, 13S, 14S, 17S)-3-hydroxy-3, 10, 13-trimethyl-1, 2, 4, 5, 6, 7, 8, 9, 11, 12, 14, 15, 16, 17-tetradecahydrocyclopenta[a]phenanthren-17-yl]ethanone) is a methyl-substituted (at the 3β position) analog of the endogenous neurosteroid allopregnanolone, a derivative of progesterone. Its empirical formula is $C_{22}H_{36}O_2$, and the molecular weight is 332.53 g/mol. The chemical structure is:

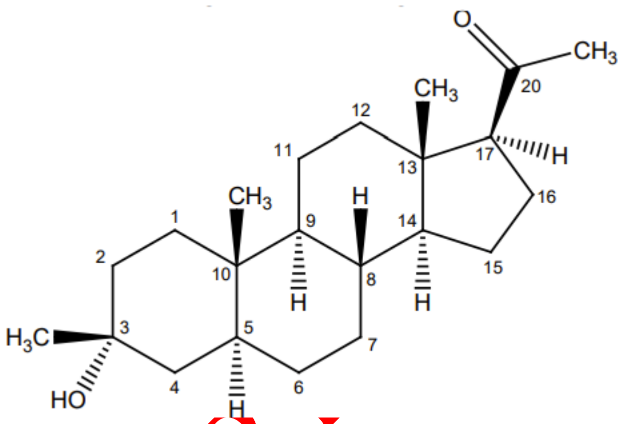

Figure 1: Ganaxolone Chemical Structure

Ganaxolone is a white to off-white crystalline powder that only exists in one crystal form and has low aqueous solubility.[7]

2.4.2.1 Primary Pharmacodynamics

Mechanism of action

Ganaxolone positively and allosterically modulates gamma-aminobutyric acid type A ($GABA_A$) receptors in the CNS by interacting with a recognition site that is distinct from other allosteric $GABA_A$ receptor modulators.

The precise mechanism by which ganaxolone exerts its therapeutic effects in the treatment of seizures associated with CDD is unknown, but its anticonvulsant effects are thought to result from this modulation of $GABA_A$ receptor function providing constant, or tonic, modulation of GABA-mediated inhibitory neurotransmission.[8]

In multiple animal seizure models, ganaxolone demonstrated a broad-spectrum anticonvulsant effect. It effectively prevented clonic and tonic seizures induced by agents such as pentylenetetrazole (PTZ), bicuculline, TBPS, aminophylline, cocaine, and by maximal electroshock (MES). Ganaxolone also inhibited fully kindled seizures in rats and mice and increased seizure threshold, indicating both suppression of seizure propagation and enhanced resistance to seizure initiation. The effective anticonvulsant doses were several times lower than those causing ataxia, suggesting a favorable protective index comparable or superior to standard antiepileptic drugs (AEDs). Additionally, ganaxolone reduced PTZ-induced behavioral abnormalities (e.g., immobility, twitches, postural changes) at the same doses that prevented seizures. Overall, these findings confirm ganaxolone's broad and potent anticonvulsant activity across various seizure types, including generalized, partial, and drug-induced seizures.[9]

A study demonstrated that pretreatment with ganaxolone (1.25–20 mg/kg, s.c.) produced a dose-dependent reduction in both behavioral and electrographic seizures in fully amygdala-kindled female mice, achieving near-complete protection at the highest dose. The effective dose for 50% seizure suppression (ED50) was 6.6 mg/kg. The anticonvulsant efficacy of ganaxolone was comparable to that of clonazepam (ED50 = 0.1 mg/kg, s.c.). Given that amygdala kindling serves as a model for mesial temporal lobe epilepsy, these findings suggest that ganaxolone may be a promising therapeutic option for managing temporal lobe seizures.[10]

This study evaluated the efficacy of ganaxolone, a synthetic neurosteroid that enhances tonic GABAA inhibition, in an animal model of cryptogenic infantile spasms (IS). Pregnant rats were primed with betamethasone (0.4 mg/kg i.p., twice on gestational day 15) to establish the IS model. On postnatal day 15 (P_{15}), the acute effects of ganaxolone were tested by administering 10, 25, or 50 mg/kg i.p. (or vehicle, β-cyclodextrin) 30 minutes before NMDA-induced spasms. To mimic human conditions, another group of rats received repeated ganaxolone treatment (20 mg/kg i.p., twice daily from P13–P15) or vehicle following NMDA-triggered spasms on P12, with additional spasms induced on P13 and P15. The latency to spasm onset and the total number of spasms during a 90-minute observation period on P15 were recorded, and behavioral tests were conducted on P19 and P21. Ganaxolone at 25 and 50 mg/kg significantly delayed spasm onset and reduced their frequency, though the highest dose caused sedation, with all rats sleeping during testing. Repeated treatment over three days similarly delayed spasm onset, reduced spasm number, and decreased exploratory behavior after multiple NMDA-induced spasms. Overall, ganaxolone effectively suppressed the development of spasms in this rat model of cryptogenic IS, suggesting its potential as a therapeutic option for treating human IS.[11]

2.4.2.2 Secondary Pharmacodynamics

This study examined whether postnatal neurosteroid-replacement therapy with ganaxolone (GNX) could reduce neurodevelopmental impairments linked to preterm birth. Time-mated sows delivered preterm (day 62) or at term (day 69), and male preterm pups were randomly assigned to receive ganaxolone (2.5 mg/kg, s.c., twice daily until term equivalence; Prem-GNX) or serve as untreated controls (Prem-CON). Surviving male juveniles underwent behavioral testing at corrected postnatal age (CPNA) day 25, and brain tissues collected at CPNA28 were analyzed for mature myelinating oligodendrocytes in the hippocampus and subcortical white matter using myelin basic protein (MBP) immunostaining. Ganaxolone treatment normalized the hyperactive behavior of preterm-born juveniles to levels comparable with term-born animals and improved MBP expression deficits in the hippocampus and subcortical white matter. However, treatment led to sedation, poor weight gain, and higher neonatal mortality. These findings indicate that ganaxolone enhances neurobehavioral outcomes in male preterm pups, though optimized dosing is needed to reduce adverse effects.[12]

2.4.2.3 Safety pharmacology

No dedicated safety pharmacology studies were identified in the public domain.

2.4.3 PHARMACOKINETICS

2.4.3.1 Absorption

Following oral administration, GNX is absorbed with a time to maximum plasma concentration (T_{max}) of 2 to 3 hours.7

2.4.3.2 Distribution

GNX exhibits extensive plasma protein binding, with about 99% bound in serum. Radiolabeled studies indicate that GNX and its metabolites distribute widely across tissues, showing tissue-to-plasma ratios exceeding five.7[, 13]

2.4.3.3 Metabolism

GNX undergoes extensive biotransformation through multiple metabolic pathways involving CYP3A4/5, CYP2B6, CYP2C19, CYP2D6, and several UGT enzymes (UGT1A3, UGT1A6, UGT1A9, UGT2B7, and UGT2B15). The primary metabolic route is via CYP3A4, producing the major metabolite 16-OH-GNX, along with several other unidentified metabolites.7[,13]

2.4.3.4 Excretion

GNX exhibits a terminal half-life of approximately 34 hours and a clearance rate of 430 L/h. Drug elimination occurs predominantly via the fecal route, with about 20% of the administered dose excreted through the kidneys.[13,14]

2.4.4 TOXICOLOGY

2.4.4.1. Single dose toxicity

GNX produced dose-related but reversible sedation in mice, rats, and dogs following acute administration, with no evidence of target organ or systemic toxicity.[13]

2.4.4.2. Repeated dose toxicity

Primary effects in animals were central nervous system clinical observations (e.g., sedation), which were dose-limiting and attributed to exaggerated pharmacology.

In the 12-month repeat-dose toxicology study in dogs, a dose-dependent increase in heart rate at ≥ 3 mg/kg/day (similar to clinical exposure levels) was observed and there were incidences of sinus tachycardia at higher doses. There were no changes in QTc intervals, blood pressure parameters, or histopathologic correlates.8

Chronic administration of GNX in mice, rats, and dogs showed no significant hematologic or clinical chemistry changes, and no organ toxicity except for increased liver and kidney weights at 40 mg/kg/day in female rats without histopathological correlates.[13]

2.4.4.3. Carcinogenicity

Oral administration of ganaxolone (150, 500, and 1000 mg/kg/day in males and 100, 300, and 1000 mg/kg/day in females) to Tg.rasH2 mice for up to 26 weeks did not result in an increase in tumors. No carcinogenicity studies have been conducted with ganaxolone in rats.7

Carcinogenicity studies were not specifically reported; however, no proliferative or preneoplastic changes were observed in repeated-dose studies up to 6 months.[13]

2.4.4.4. Mutagenesis

Ganaxolone was negative for genotoxicity in *in vitro* (Ames and mouse lymphoma) and *in vivo* (rat bone marrow micronucleus) assays. The major circulating human metabolite, oxy-dehydroganaxolone, was negative for mutagenicity in the *in vitro* Ames assay but positive for clastogenicity in an *in vitro* mammalian chromosomal aberration test in human peripheral blood lymphocytes.7

GNX was non-mutagenic in the mouse lymphoma assay, *Salmonella–E. coli*/mammalian-microsome reverse mutation assay, and rat *in vivo* micronucleus assay.[13]

2.4.4.5. Reproductive and Developmental Toxicity

Oral administration of ganaxolone (0, 10, 20 or 40 mg/kg/day) to male and female rats prior to and throughout mating and continuing in females during early gestation resulted in alterations in estrous cyclicity at the high dose. There were no effects on spermatogenesis, reproductive performance and fertility, or early embryonic development. The highest dose tested (40 mg/kg/day) was associated with plasma exposures (AUC) less than that in adult humans at the maximum recommended human dose of 1800 mg.7

In rats, GNX did not affect sperm count, motility, mating performance, fertility, fetal implantation, or postnatal viability and development, and 6-week neonatal exposure did not alter developmental milestones or behavior.[13]

2.4.5 INTEGRATED OVERVIEW AND CONCLUSIONS

Seizure disorders are neurological conditions marked by recurrent seizures that can result in significant disability and a reduced quality of life. Ganaxolone is a synthetic neuroactive steroid with low aqueous solubility and is a methyl analogue of the endogenous neurosteroid allopregnanolone. It functions as a positive allosteric modulator of gamma-aminobutyric acid type A ($GABA_A$) receptors in the CNS by interacting with a recognition site that is distinct from other allosteric $GABA_A$ receptor modulators.

The precise mechanism by which ganaxolone exerts its therapeutic effects in the treatment of seizures associated with CDD is unknown, but its anticonvulsant effects are thought to result from this modulation of $GABA_A$ receptor function providing constant, or tonic, modulation of GABA-mediated inhibitory neurotransmission.

The effective dose required to achieve 50% seizure suppression (ED50) was 6.6 mg/kg. Ganaxolone demonstrated activity in reducing temporal lobe seizures, indicating its potential utility in this setting. In a rat model of cryptogenic infantile spasms, ganaxolone also suppressed the development of spasms, supporting its possible role as a therapeutic option for human infantile spasms.

After oral administration, GNX reaches peak plasma levels in 2–3 hours and is highly protein bound (~99%). Radiolabeled studies show broad tissue distribution with tissue-to-plasma ratios above five. GNX undergoes extensive metabolism *via* CYP3A4/5, CYP2B6, CYP2C19, CYP2D6, and multiple UGT enzymes, with CYP3A4 producing the major metabolite 16-OH-GNX. Its terminal half-life is about 34 hours, and clearance is approximately 430 L/h. Elimination occurs mainly in feces, with around 20% excreted in urine.

GNX induced dose-dependent, reversible sedation in mice, rats, and dogs after acute dosing, with no evidence of target organ or systemic toxicity.

In a 12-month repeat-dose study in dogs, GNX produced a dose-related increase in heart rate at doses ≥ 3 mg/kg/day, comparable to clinical exposure levels, with sinus tachycardia observed at higher doses. No effects were noted on QTc intervals, blood pressure, or associated histopathology. Chronic GNX administration in mice, rats, and dogs showed no significant hematologic or clinical chemistry changes, and no organ toxicity except for increased liver and kidney weights at 40 mg/kg/day in female rats without histopathological correlates.

No carcinogenicity, mutagenicity, or reproductive and developmental toxicity findings have been reported.

Evidence from the literature data indicated that ganaxolone oral suspension (50 mg/mL) was safe, well tolerated, and effective in achieving its intended therapeutic outcomes.

2.4.6 LITERATURE REFERENCES

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